Answer Partitions, Oracle Response Spectra, and Exact Quantum Query Advantage

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An oracle problem has two parts: an answer partition, which says which hidden instances must receive the same answer, and an access model, which says what a query returns. We show that the access model can change the exact quantum query complexity in ways no classical query can detect. One query solves a task exactly if and only if some probe makes the output states of different answer cells orthogonal; we state this criterion together with a certificate that proves when it fails. For an oracle that applies a response unitary V at the queried address, one query solves a dictator (a function equal to a single input bit) exactly if and only if 0 lies in the convex hull of the spectrum of V, and a two-bit parity, including Deutsch's problem, if and only if -1 is an eigenvalue of V; no other total Boolean function is one-query exact. Hence a qutrit transposition and a cyclic shift, which return the same values on a blank register, need one and two exact queries; with bounded error, one query to the shift still succeeds with the optimal probability 7/8. An inaccessible copy of k address bits raises the cost of Bernstein–Vazirani identification from one query to $\min\{k+1, n\}$, although every classical value stays the same. For the standard oracle we reorganize the known exact complexities of the 35 balanced three-bit partitions and classify all 6435 balanced four-bit partitions exactly; rational dual certificates prove the 6040 four-bit lower bounds that were known only from numerical semidefinite programming.

I. INTRODUCTION

Deciding whether a one-bit function is constant or balanced takes one query to the standard XOR oracle on a quantum computer and two on a classical one [1–3]. A qutrit oracle that returns exactly the same classical values, the cyclic shift $y \mapsto y+b_x \bmod 3$, needs two quantum queries for an exact answer, whereas a qutrit transposition that agrees with it on a blank register needs only one (Sec. III). The difference lies in the spectrum of the oracle's response, which no classical query can see.

In general, a query problem has two parts: an answer partition, which groups the hidden instances into cells whose members share the correct answer, and an access model, which says what each query returns. Query complexity, and with it any quantum speedup, belongs to this pair, not to the task alone. That it depends on the access model is known: simulating an in-place oracle for a permutation of N points can require $\Theta(\sqrt{N})$ uses of the standard oracle for the same function [4]; the comparison of quantum and classical oracle power depends on how the oracle is represented [5]; a richer classical oracle solves Deutsch's problem with one query [6]; and a faulty oracle removes Grover's speedup [7].

Earlier work, including the author's, treated quantum decision problems as partitions or identifications of prepared quantum states [8–13]. In the query model the partition is instead defined on black-box transformations, and the oracle and the protocol must first turn it into orthogonal output states. This paper asks which features of the oracle, beyond the classical values it returns, decide how many queries this takes.

This happens in four steps, used throughout the paper and listed in Sec. VI. First, the task fixes the answer partition: instances with the same correct answer share a cell, and only the cell must be learned. Second, the access model fixes what one oracle call does. Third, an instance-independent probe, typically a superposition fanned out from a single basis state, is sent through the oracle, which turns it into an output state that depends on the hidden instance. Fourth, one call solves the task exactly if and only if some probe makes the output states of different cells orthogonal; overlaps within a cell do no harm. In the example above, this orthogonality needs a relative factor -1 between interfering branches, which the flip and the transposition supply and the cyclic shift does not.

The main new result is a spectral criterion (Theorem 1 and Proposition 4). It concerns dictators and parities. A dictator is a function whose value is just one of its input bits, $f(b) = b_i$ (or its negation): one bit dictates the answer and the others do not matter, so classically one query, to address i, suffices. For an oracle that applies a response unitary V at the queried address, one query solves a dictator exactly if and only if 0 lies in the convex hull of the spectrum of V, and a two-bit parity, including Deutsch's problem, if and only if -1 is an eigenvalue of V; no other total Boolean function is one-query exact. For the XOR oracle this reduces to the characterization by Chen, Ye, and Li [14]. It follows that two reversible completions of the same blank-register table can have different exact costs, that a phase invisible to basis queries can restore the advantage, and that an odd cyclic response keeps a bounded-error advantage although it loses the exact one (Sec. III). The second new result is an exact classification of all balanced four-bit tasks for the standard oracle: rational dual certificates prove the two-query lower bounds that were previously known only numerically [15] (Sec. IV). Section II sets up

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the framework, a block form of standard discrimination conditions and its dual. Section V treats an oracle that leaves an inaccessible copy of the address; this changes Bernstein–Vazirani costs although every classical value stays the same. Table I separates prior results from new ones.

These results also correct a common picture of quantum parallelism: what matters for the separations below is how the oracle responds (its phases, spectrum, accessible registers, and environment), not how many branches a superposition has.

II. ANSWER PARTITIONS AND EXACT ONE-QUERY RESOLUTION

Let Ω be a finite set of instances and let $g : \Omega \to J$ give the correct answer. The task defines the *answer partition*

$$\Pi_g = \{C_j : j \in g(\Omega)\}, \qquad C_j = g^{-1}(j). \tag{1}$$

The partition says what must be found, not how. In the query model the hidden instance labels a black-box transformation; the algorithm is not given a state $|a\rangle$ that it could simply measure. The question is whether the given oracle and a fixed protocol can bring the cell label into a register that can be measured.

Let $\rho_a^{(T)}$ be the state of the measurable registers after T queries on instance a. Exact resolution means that one instance-independent measurement $\{P_j\}$ satisfies

$$P_j P_\ell = \delta_{j\ell} P_j, \quad \sum_j P_j = I, \quad \mathrm{Tr}(P_{g(a)} \rho_a^{(T)}) = 1. \tag{2}$$

States from different cells then have orthogonal supports; states within a cell need not be orthogonal. The cells C_j are sets of classical instances, not subspaces; the subspaces $P_j \mathcal{H}_{\mathrm{out}}$ exist only after an oracle and a protocol have produced them. The partition thus prescribes which output overlaps must vanish; it is not a ready-made observable.

$Q_\epsilon(\Pi;\mathcal{U})$ and $R_\epsilon(\Pi;\mathcal{U}_{\mathrm{cl}})$ denote the smallest worst-case numbers of quantum and classical calls that achieve worst-case error at most ϵ, with matching access on both sides; $Q_E \equiv Q_0$ is the exact quantum complexity, and D is the worst-case number of queries of an exact deterministic classical algorithm (the depth of an optimal decision tree). With worst-case counting, $D = R_0$: fixing the random choices of a zero-error randomized protocol that never uses more than T queries leaves a deterministic protocol that is always correct with at most T queries. (Counted in expectation, as is common, zero-error cost can be smaller than D.) A deterministic classical protocol groups the instances by the transcript of answers they produce; it succeeds exactly only if instances with the same transcript always lie in the same cell. A quantum protocol keeps no such intermediate record; it changes the overlaps among the instance states and measures only after the cells have separated. A speedup is a gap between the quantum and the classical cost for the same task and the same access.

All Hilbert spaces are finite dimensional. For a family of unitary oracles $\{U_a : a \in \Omega\}$, a general one-query protocol prepares a pure state $|\Psi\rangle$ of the query system Q and a reference system R, applies $U_a \otimes I_R$, and then applies a fixed channel and measurement. Write

$$|\Psi_a\rangle = (U_a \otimes I_R)\,|\Psi\rangle\,, \qquad \sigma = \mathrm{Tr}_R\,|\Psi\rangle\langle\Psi|\,. \tag{3}$$

The Gram matrix of the output states is

$$G_{aa'} = \langle\Psi_a|\Psi_{a'}\rangle = \mathrm{Tr}\big(\sigma U_a^\dagger U_{a'}\big)\,. \tag{4}$$

Proposition 1 (Block discrimination)**.** *One query to $\{U_a\}$, with any reference system allowed, resolves Π_g exactly if and only if some density operator σ satisfies*

$$\mathrm{Tr}\big(\sigma U_a^\dagger U_{a'}\big) = 0 \quad \textit{whenever} \quad g(a) \neq g(a'). \tag{5}$$

Proof. Exact discrimination requires the output states of different cells to span orthogonal subspaces, that is, Eq. (5) for every pair from different cells; no fixed channel can make nonorthogonal states perfectly distinguishable. Conversely, orthogonal subspaces are distinguished by the projectors onto them. By purification, optimizing over σ covers mixed probes and arbitrary ancillas. □

Equation (5) is the block form of the standard conditions for perfect discrimination of unitaries [16, 17]; it is not a new theorem about channel discrimination. Here it links an answer partition to a given oracle family. The Gram matrix need only be block diagonal with respect to Π_g; states within one cell may overlap or coincide. Identifying the instance completely is the special case of one-element cells, where G must be diagonal. Since zero queries cannot resolve a nontrivial partition, Eq. (5) characterizes $Q_E(\Pi_g;\mathcal{U}) = 1$. With bounded error, orthogonality is replaced by a measurement that reaches the required success probability. Lower bounds for several queries come from adversary methods [24], whose general form characterizes state conversion and function evaluation [25, 26].

When Eq. (5) has no solution, a single matrix proves it.

Proposition 2 (Dual certificate)**.** *Let $X = \{(a, a') : g(a) \neq g(a')\}$. Exactly one of the following holds: (i) a density operator σ satisfies Eq. (5); (ii) there is a complex matrix Γ, supported on X and satisfying $\Gamma_{a'a} = \Gamma^*_{aa'}$, such that*

$$Z(\Gamma) = \sum_{(a,a')\in X} \Gamma_{aa'}\, U_a^\dagger U_{a'} \succ 0. \tag{6}$$

Proof. $Z(\Gamma)$ is Hermitian because $(U_a^\dagger U_{a'})^\dagger = U_{a'}^\dagger U_a$. If (i) and (ii) both held, then $\mathrm{Tr}[\sigma Z(\Gamma)] = 0$, contradicting $Z(\Gamma) \succ 0$. If (ii) fails, the real subspace $\{Z(\Gamma)\}$ of Hermitian operators misses the open convex cone of positive definite operators, so a separating hyperplane gives

TABLE I. Relation to prior work. "Known" lists results available before this paper; "This work" lists what is added here.

Topic	Known	This work
Exact one-query criterion	Perfect discrimination of unitaries [16, 17]	Block form for answer partitions and its dual certificate (Props. 1 and 2); standard convex duality, stated for use below
One-query exact class	XOR oracle: dictators and two-bit parities [14]; partial functions [18]	Every response unitary V: dictators iff $0 \in \operatorname{conv}(\operatorname{spec} V)$, two-bit parities iff $-1 \in \operatorname{spec} V$, no other total Boolean functions (Thm. 1, Prop. 4)
Dependence on the oracle model	Oracle representation [5]; in-place versus standard oracles [4]; de-quantised Deutsch oracle [6]; faulty oracles [7]	A spectrum invisible to classical queries decides the exact cost: two completions of one blank-register table, phase-dressed shift (Cor. 1); sign oracles cannot separate complementary truth tables (Prop. 3)
Bounded error, odd cyclic response	—	Optimal unambiguous one-query solution, worst-case success $(3+\cos\frac{\pi}{d})/4$ (Prop. 5)
Parity	$Q_E(\mathrm{Par}_N) = \lceil N/2 \rceil$ [19, 20]	Degree bound for every response; open qutrit question (Sec. III)
Three-bit functions	Explicit optimal exact algorithms for all functions [15]	No new values; reorganization as balanced partitions (Table II)
Four-bit functions	Numerical SDP values for all functions [15], used in structural work [21]	Exact rational dual certificates for all 6435 balanced partitions (Table III)
Decohered access	Which-way information [22]; coherence in Bernstein–Vazirani [23]	Exact costs with an inaccessible witness: surviving partition in one query versus $n-k$; identification in exactly $\min\{k+1, n\}$ queries (Prop. 6)

a nonzero σ with $\mathrm{Tr}(\sigma P) \geq 0$ for all $P \succ 0$, that is $\sigma \succeq 0$, and $\mathrm{Tr}[\sigma Z(\Gamma)] = 0$ for all admissible Γ. Taking Γ supported on a single pair (a, a') and its transpose, with real and with imaginary entries, gives $\mathrm{Tr}(\sigma U_a^\dagger U_{a'}) = 0$; normalizing σ gives (i). □

Like an adversary matrix, Γ weights only pairs from different cells. After t queries the same weights define the progress measure $\sum_{(a,a')\in X} \Gamma_{aa'} \langle \psi_a^{(t)} | \psi_{a'}^{(t)} \rangle$ of the adversary method [24], and the query cost measures how fast such weighted cross-cell overlaps can be driven to zero; this is what it means for a quantum algorithm to "separate" answer cells. When all $U_a^\dagger U_{a'}$ commute, as for the point oracles below, which are diagonal in the eigenbasis of their response, only the diagonal of σ in a common eigenbasis enters Eq. (5), and the criterion becomes a linear feasibility problem for a probability vector.

The criterion also separates a mere change of representation from a change of access. If

$$V_a = A U_a B \tag{7}$$

for fixed unitaries A, B, then $V_a^\dagger V_{a'} = B^\dagger U_a^\dagger U_{a'} B$, so both families reach the same Gram matrices. More generally, families that simulate each other exactly with $O(1)$ calls have the same quantum query complexity up to a constant factor; an approximate simulation adds its error to ϵ. For a quantum–classical gap to carry over, the classical oracles must simulate each other in the same way. Families not related by Eq. (7) can differ in their exact one-query cost, as the next section shows.

III. THE RESPONSE SPECTRUM DECIDES EXACT ONE-QUERY POWER

A. Deutsch's problem and the sign oracle

Let $b = (b_0, b_1)$ be the truth table of a Boolean function and use

$$O_b |x, y\rangle = |x, y \oplus b_x\rangle . \tag{8}$$

With the address in $|+\rangle$ and the answer qubit in $|-\rangle$, phase kickback leaves the address register in the states (the answer qubit factors out)

$$|\psi_{00}\rangle = -|\psi_{11}\rangle = |+\rangle , \quad |\psi_{01}\rangle = -|\psi_{10}\rangle = |-\rangle . \tag{9}$$

Constant and balanced functions give orthogonal states, while the two functions within each cell give the same state up to sign. The Gram matrix is block diagonal but not diagonal, and the result is the familiar separation of one quantum versus two classical queries [1–3].

Hadamard gates on the answer qubit turn Eq. (8) into the controlled phase $|x, z\rangle \mapsto (-1)^{z b_x} |x, z\rangle$, so the two forms are equivalent in the sense of Eq. (7). The sign oracle $P_b |x\rangle = (-1)^{b_x} |x\rangle$, without the control qubit z, is weaker. A basis query to it reveals nothing about b_x, and

for the complemented truth table $\bar{b}$ one has $P_{\bar{b}} = -P_b$, so the two differ only by a global phase.

Proposition 3 (Complement obstruction)**.** *With access to $\{P_b\}$ alone (any ancillas, but no control on P_b), no number of queries resolves a partition that puts some b and its complement $\bar{b}$ in different cells, even with error $\epsilon < 1/2$.*

Proof. Each use of $P_{\bar{b}} \otimes I$ equals $-(P_b \otimes I)$, and every branch of an adaptive protocol picks up the same sign. The final density operators for b and $\bar{b}$ are therefore identical, and so are the output distributions. □

Deutsch's constant–balanced partition is closed under complement and is still solved by one query to P_b. Three-bit majority, the parity of an odd number of bits, and every other self-dual function cannot be resolved at all. Of the balanced partitions in Sec. IV, only 3 of 35 for three bits (the two-bit parities) and 35 of 6435 for four bits are closed under complement; the sign oracle resolves none of the others, while the XOR oracle resolves all of them with at most two ($N = 3$) or three ($N = 4$) queries. A coherent control on P_b would restore the missing relative phase, but control is extra black-box access, not something one can add to an unknown unitary [27].

B. Response unitaries

Let V be any unitary and consider the family, indexed by the same four truth tables, that applies V to the target whenever $b_x = 1$:

$$U_b^{(V)} = \sum_{x=0}^{1} |x\rangle\langle x| \otimes V^{b_x}. \tag{10}$$

Access conventions. V is known. Between queries the algorithm may apply any fixed operation, including preparing eigenvectors of V, at no query cost; it may not control, invert, or otherwise modify the oracle unless that is explicitly allowed. The phase of V matters: V^{b_x} acts only on the branch selected by the address, so replacing V by $e^{i\theta}V$ multiplies Eq. (10) by $e^{i\theta b_x}$, a phase that differs between address branches and is therefore observable; it can change the spectral condition below (Corollary 1). This differs from access to an unknown unitary or channel, which is defined only up to a global phase.

Theorem 1 (Response spectrum)**.** *One query to $\{U_b^{(V)}\}$, with any probe and reference, solves Deutsch's problem exactly if and only if -1 is an eigenvalue of V.*

Proof. If $V|w\rangle = -|w\rangle$, the probe $|+\rangle|w\rangle$ gives ordinary phase kickback and Eq. (9). Conversely, give weight $\Gamma = 1$ to the four cross-cell pairs (and their transposes) in Proposition 2. The pairs $(00, 01)$, $(00, 10)$, $(11, 01)$, and $(11, 10)$ contribute $I \oplus V$, $V \oplus I$, $V^\dagger \oplus I$, and $I \oplus V^\dagger$ on the address blocks $x = 0, 1$, so

$$Z = 2\, I_2 \otimes \left(2I + V + V^\dagger\right) = 2\, I_2 \otimes (I+V)(I+V)^\dagger. \tag{11}$$

Z is positive definite unless $I + V$ is singular, that is, unless -1 is an eigenvalue of V. □

In words: at each of the two addresses the probe sees an average of eigenvalues of V, a point of $\mathrm{conv}(\mathrm{spec}\, V)$, and orthogonality across cells forces both points to -1. Since -1 lies on the unit circle, an average of eigenvalues equals -1 only if -1 is itself an eigenvalue. Qudit versions of Deutsch–Jozsa enlarge the address and response systems [2, 3, 28]; Theorem 1 instead keeps the four Deutsch instances fixed and varies the response unitary. It also differs from "useless-query" bounds, which derive quantum lower bounds, under a prior distribution, from classical queries that reveal nothing about a property [29].

Corollary 1 (Reversible completions)**.** *On a qutrit, let T be the transposition $|0\rangle \leftrightarrow |1\rangle$ that fixes $|2\rangle$, let $X_3|y\rangle = |y+1 \bmod 3\rangle$ be the cyclic shift, and let $X_3' = e^{i\pi/3}X_3$. All three map the blank register $|0\rangle$ to $|1\rangle$ up to a phase ($X_3'|0\rangle = e^{i\pi/3}|1\rangle$), so measuring a blank target returns b_x in each case; X_3 and X_3' moreover agree on every computational-basis input up to a phase. Their spectra are $\{1, 1, -1\}$, $\{1, e^{2\pi i/3}, e^{4\pi i/3}\}$, and $\{e^{i\pi/3}, -1, e^{-i\pi/3}\}$, so the exact quantum cost of Deutsch's problem is $Q_E = 1$, 2, and 1, respectively, while $D = 2$ in all three cases. More generally, for $X_d|y\rangle = |y+1 \bmod d\rangle$ with $d \geq 2$,*

$$Q_E^{(d)} = \begin{cases} 1, & d \text{ even}, \\ 2, & d \text{ odd}, \end{cases} \qquad D^{(d)} = 2. \tag{12}$$

Specifying an oracle by what it does to a blank target, $|x\rangle|0\rangle \mapsto |x\rangle|b_x\rangle$, fixes only one column of each V^{b_x}. How the rest is filled in, by a transposition or a cyclic shift, decides whether Deutsch's exact advantage exists, and so does a phase that no basis query can see: the factor $e^{i\pi b_x/3}$ in $X_3'^{b_x}$ differs between address branches; it is a phase oracle composed with the shift. Neither the classical values nor even the full action on basis states up to phases determines Q_E. The effect is a constant factor, not an asymptotic separation.

C. All point oracles with a response unitary

The same argument works for any number of bits. Let $f : \{0,1\}^N \to \{0,1\}$ be nonconstant, with oracle $U_b^{(V)} = \sum_{i=1}^{N} |i\rangle\langle i| \otimes V^{b_i}$.

Proposition 4 (One-query class)**.** *One query to $\{U_b^{(V)}\}$ resolves a nonconstant f exactly if and only if, up to negation, either $f = b_i$ and $0 \in \mathrm{conv}(\mathrm{spec}\, V)$, or $f = b_i \oplus b_j$ and $-1 \in \mathrm{spec}\, V$. The dictator condition $0 \in \mathrm{conv}(\mathrm{spec}\, V)$ holds in particular if $V|y\rangle \perp |y\rangle$ for some basis state $|y\rangle$.*

Proof. Let $\rho_i \succeq 0$ be the diagonal block of σ at address i and $p_i = \mathrm{Tr}\,\rho_i$, so $\sum_i p_i = 1$. Since $U_b^\dagger U_{b'} = \sum_i |i\rangle\langle i| \otimes$

$V^{b'_i - b_i}$, only the ρ_i enter Eq. (5). If f depends on b_i, some b has $f(b) \neq f(b \oplus e_i)$, and Eq. (5) reads $1 - p_i + \mathrm{Tr}(\rho_i V^{\pm 1}) = 0$. Since $|\mathrm{Tr}(\rho_i V^{\pm 1})| \leq p_i$, this forces $p_i \geq 1/2$, so f depends on at most two bits. If it depends on two, i and j, then $p_i = p_j = 1/2$ and $\mathrm{Tr}(\rho_i V^{\pm 1}) = -p_i$, which places ρ_i and ρ_j in the -1 eigenspace of V. The overlap of the outputs for b and $b \oplus e_i \oplus e_j$ is then $-1 \neq 0$, so these instances share a cell, $f(b) = f(b \oplus e_i \oplus e_j)$, and f is $b_i \oplus b_j$ or its negation. A -1 eigenvector gives the Deutsch construction on addresses i, j. For a dictator b_i, the cross-cell pairs $(0^N, e_i)$ and $(0^N, e_i \oplus e_j)$, $j \neq i$, give $\mathrm{Tr}(\rho_i V) + \sum_{j \neq i} p_j = 0$ and $\mathrm{Tr}(\rho_j V) = p_j$, so the density operator $\tau = \sum_j \rho_j$ has $\mathrm{Tr}(\tau V) = 0$, that is, $0 \in \mathrm{conv}(\mathrm{spec}\, V)$. Conversely, the probe $|i\rangle\,|w\rangle$ with $\langle w|V|w\rangle = 0$ makes every cross-cell overlap vanish. With an additional null address, on which the oracle acts as the identity (as in the phase form used in Sec. IV), the same argument holds with $\mathrm{spec}\, V$ replaced by $\mathrm{spec}\, V \cup \{1\}$ in the dictator condition. □

For a dictator b_i it suffices to query address i alone; one query then succeeds exactly when some target state $|w\rangle$ is turned into an orthogonal one, $\langle w|V|w\rangle = 0$, that is, when an average of eigenvalues of V vanishes. For $V = -I$, the family is the sign oracle of Proposition 3: one query resolves two-bit parities, but no number of queries resolves a dictator.

For the XOR oracle, $V = X$, Proposition 4 reproduces the characterization of exact one-query algorithms by Chen, Ye, and Li [14]. Proposition 4 concerns total functions. With a promise, more tasks can be one-query exact, the Deutsch–Jozsa problem [2] being the standard example; partial functions are treated in Ref. [18]. For an odd cyclic response X_d only the dictators remain, and a single classical query already resolves them.

D. Bounded error for odd cyclic responses

Corollary 1 concerns exact solutions. With bounded error, the odd cyclic response keeps a quantum advantage. As a referee pointed out, a single eigenvector of X_3 with eigenvalue $e^{2\pi i/3}$, the address in $|+\rangle$, and a measurement of the address in $\{|+\rangle, |-\rangle\}$ already give success 1 on constant and 3/4 on balanced instances, whereas one classical query cannot exceed worst-case success 1/2. Spreading the probe over the two eigenvalues adjacent to -1 raises the worst case further and makes every conclusive answer correct.

Proposition 5 (Optimal one-query solution)**.** *For odd $d \geq 3$, one query to $\{U_b^{(X_d)}\}$ solves Deutsch's problem with zero error and a conclusive answer with probability $(1 + \cos\frac{\pi}{d})/2$ on every instance. Guessing on the inconclusive outcome gives worst-case success $(3 + \cos\frac{\pi}{d})/4$, that is 3/4 conclusive and 7/8 overall for $d = 3$, and no one-query protocol does better. One classical query never gives a conclusive zero-error answer and has worst-case success 1/2.*

Proof. Let $|v_\pm\rangle$ be eigenvectors of X_d with the eigenvalues $\lambda_\pm = -e^{\pm i\pi/d}$ adjacent to -1, and use the probe $\frac{1}{2}\sum_{x=0}^{1}\sum_\pm |x\rangle\,|v_\pm\rangle$. Writing $\gamma = \cos\frac{\pi}{d}$ and using $(\lambda_+^k + \lambda_-^k)/2 = -\gamma$ for $k = \pm 1$, the output Gram matrix is $G_{bb'} = \frac{1}{2}\sum_x c_x$ with $c_x = 1$ if $b_x = b'_x$ and $c_x = -\gamma$ otherwise. Hence $|\psi_{00}\rangle + |\psi_{11}\rangle = |\psi_{01}\rangle + |\psi_{10}\rangle$, with squared norm $2(1-\gamma)$, while $|\psi_{00}\rangle - |\psi_{11}\rangle$ and $|\psi_{01}\rangle - |\psi_{10}\rangle$ are orthogonal to each other and to that common vector, each with squared norm $2(1+\gamma)$. Projecting onto these three directions returns "constant," "balanced," or "inconclusive"; each instance gives the correct conclusive outcome with probability $(1+\gamma)/2$ and never the wrong one.

For optimality, write any pure probe, including a reference, as $|0\rangle\,|A\rangle + |1\rangle\,|B\rangle$ with $\|A\|^2 + \|B\|^2 = 1$, and put $W = V \otimes I_R$ with $V = X_d$. Let ρ_C and ρ_B be the outputs averaged uniformly over the constant and over the balanced instances. Their diagonal address blocks coincide, and their off-diagonal blocks differ by $\frac{1}{2}\,|(I-W)A\rangle\langle(I-W)B|$, so $\|\rho_C - \rho_B\|_1 = \|(I-W)A\|\,\|(I-W)B\| \leq \frac{1}{2}\|I - V\|^2$. Worst-case success cannot exceed success under this prior, which by the Helstrom bound is at most $\frac{1}{2} + \frac{1}{8}\|I - V\|^2$. For odd d, $\|I - X_d\|^2 = 2 + 2\cos\frac{\pi}{d}$, which gives $(3 + \cos\frac{\pi}{d})/4$. Classically, one query reveals one bit b_x, and both cells remain consistent with every transcript. □

The bound also makes the construction optimal among unambiguous protocols. If every instance gave a conclusive answer with probability at least p, guessing on the inconclusive outcome would give worst-case success at least $(1+p)/2$; hence $p \leq (1 + \cos\frac{\pi}{d})/2$, which the construction attains.

The bound $\frac{1}{2} + \frac{1}{8}\|I - V\|^2$ holds for every response V and reaches 1 only if $\|I - V\| = 2$, that is, only if -1 is an eigenvalue; this is a second proof of the necessity part of Theorem 1. The construction parallels Deutsch's original algorithm, which also answered conclusively only part of the time [1]; here the inconclusive probability $(1 - \cos\frac{\pi}{d})/2$ is set by how close the spectrum comes to -1, and it vanishes as $d \to \infty$.

E. Parity

Proposition 4 shows that one query resolves the parity of N bits only for $N \leq 2$. Proposition 2 gives an explicit witness for the XOR oracle: with $\Gamma = 1$ on the pairs $(0^N, e_i)$,

$$Z = \sum_{i=1}^{N} \left(O_{e_i} + O_{e_i}^\dagger\right) = 2N\, I - 4\Pi_-, \tag{13}$$

where $\Pi_- = \sum_i |i\rangle\langle i| \otimes |-\rangle\langle -|$; Z is positive definite exactly when $N > 2$. For more queries, note that $U_b^{(V)} = \sum_i |i\rangle\langle i| \otimes [I + b_i(V - I)]$ is affine in b for every response V. The amplitudes after T queries are therefore polynomials of degree at most T in the input bits, the acceptance

probability has degree at most $2T$, and parity has exact degree N. Thus $T \geq N/2$ for every response, and for the XOR oracle applying the Deutsch construction to disjoint pairs gives the matching algorithm, so

$$Q_E(\mathrm{Par}_N) = \lceil N/2 \rceil, \qquad D(\mathrm{Par}_N) = N \tag{14}$$

for point-value access [19, 20]. For the qutrit shift X_3, Theorem 1 forbids the pair construction, and $\lceil N/2 \rceil \leq Q_E(\mathrm{Par}_N; X_3) \leq N$. Whether the upper bound is attained is open; if it is, two oracles with identical classical point values would differ by a factor of two for all N.

If instead an oracle $\widetilde{O}_b\,|y\rangle = |y \oplus \mathrm{Par}(b)\rangle$ that returns the parity counts as one call, parity takes one query, classically and quantum mechanically. This does not contradict Eq. (14); such an oracle already returns the answer.

IV. EXACT CENSUS OF BALANCED POINT-ORACLE TASKS

So far the access varied and the task stayed fixed. Now the standard oracle stays fixed and the task varies. The hidden instance is $b \in \{0,1\}^N$, accessed through Eq. (8) with N addresses, and we consider every two-valued answer with 2^{N-1} instances in each cell. Swapping the two answers gives the same partition, so the number of such partitions is

$$M_N = \frac{1}{2}\binom{2^N}{2^{N-1}}. \tag{15}$$

Permuting coordinates, complementing coordinates, and swapping the answers preserve both D and Q_E. These symmetries reduce the $M_3 = 35$ partitions to 6 orbits and the $M_4 = 6435$ partitions to 58. The counts below include the orbit sizes.

Methods. D was computed exactly by dynamic programming over subcubes. By Proposition 4 (Ref. [14]), $Q_E = 1$ exactly for dictators and two-bit parities. Upper bounds on Q_E come from adaptive trees whose nodes read either one bit or the parity of two bits, at one query each; such trees prove upper bounds but do not capture all exact quantum algorithms [15]. Lower bounds $Q_E \geq 3$ come from the two-query semidefinite program of Barnum, Saks, and Szegedy [30], in its bounded-error form, which is numerically better behaved than an exact feasibility test. In the phase form of the point oracle, $|i\rangle \mapsto (-1)^{b_i}|i\rangle$ with a null address $i = 0$ and $b_0 = 0$, put $D_i = \mathrm{diag}_b\big((-1)^{b_i}\big)$ (so $D_0 = I$). A two-query algorithm with worst-case success p yields positive semidefinite $2^N \times 2^N$ matrices with $\sum_i A_i = J$ (the all-ones matrix), $\sum_i D_i A_i D_i = \sum_i B_i$, $\sum_i D_i B_i D_i = \Gamma_0 + \Gamma_1$, and $(\Gamma_{f(b)})_{bb} \geq p$. If real symmetric Y_1, Y_2, Y_3 and a probability vector λ satisfy

$$\begin{aligned} &Y_1 \succeq D_i Y_2 D_i, \quad Y_2 \succeq D_i Y_3 D_i \quad (i = 0, \ldots, N), \\ &Y_3 \succeq \Lambda_j \quad (j = 0, 1), \end{aligned} \tag{16}$$

TABLE II. Balanced three-bit tasks up to coordinate permutations, input complements, and answer exchange. The orbit sizes sum to 35. Here $\mathrm{Sel}(c, u, v) = u$ for $c = 0$ and v for $c = 1$. The values of Q_E are not new: Ref. [15] gives explicit optimal exact algorithms for every function on up to three bits.

Representative	Orbit	D	Q_E
b_1	3	1	1
$b_1 \oplus b_2$	3	2	1
Par_3	1	3	2
Maj_3	4	3	2
$b_1 \oplus b_2 b_3$	12	3	2
$\mathrm{Sel}(b_1, b_2, b_3)$	12	2	2

with $\Lambda_j = \mathrm{diag}_b\big(\lambda_b[f(b) = j]\big)$, then

$$\begin{aligned} \mathbf{1}^{\mathsf{T}} Y_1 \mathbf{1} &= \textstyle\sum_i \langle Y_1, A_i\rangle \geq \langle Y_2, \sum_i B_i\rangle \geq \sum_j \langle Y_3, \Gamma_j\rangle \\ &\geq \textstyle\sum_b \lambda_b (\Gamma_{f(b)})_{bb} \geq p, \end{aligned} \tag{17}$$

so $\mathbf{1}^{\mathsf{T}} Y_1 \mathbf{1} < 1$ proves $Q_E \geq 3$. We solved Eq. (16) numerically with a barrier method, added 3ε, 2ε, and ε times the identity to Y_1, Y_2, and Y_3 with $\varepsilon = 2^{-20}$, which raises every constraint matrix in Eq. (16) by εI, rounded every entry to a dyadic rational with denominator 2^{40}, and verified all twelve matrix inequalities and $\lambda \geq 0$, $\sum_b \lambda_b = 1$ in exact integer arithmetic (positive pivots in Gaussian elimination). Each certificate is a list of integers, given in the Supplemental Material, and can be checked without the solver.

Use of AI tools. The census code and the Lean formalization were written with an OpenAI Codex assistant, and the dual-certificate solver and the exact verifier with Anthropic's Claude (versions are given in the Acknowledgments). No reported result rests on these tools alone: the census counts were reproduced by two independently written programs and agree with Ref. [15], the Lean audits reproduce the finite census, and every certificate is checked by a verifier that uses only exact integer arithmetic and does not depend on the solver.

Three bits. For $N = 3$ the lower bounds from the one-query class match the explicit trees for every task (Table II). These values are known: Montanaro, Jozsa, and Mitchison give optimal exact algorithms for every three-bit function [15]; we only arrange them as balanced partitions. Of the 35 balanced tasks, 20 have an exact advantage: three by a factor of two and seventeen by a factor of 3/2. The trees show how such algorithms work. For three-bit majority, first obtain $b_1 \oplus b_2$; if it is 0, read b_1, otherwise read b_3; this is the algorithm of Ref. [15], Sec. 6.2. Likewise, for $F = b_1 \oplus b_2 b_3$, first read b_2, then b_1 if $b_2 = 0$, or the Deutsch parity $b_1 \oplus b_3$ if $b_2 = 1$.

Four bits. For $N = 4$ the one-query class and explicit two-query trees settle 15 orbits covering 395 partitions. For each of the remaining 43 orbit representatives, covering 6040 partitions, explicit trees give $Q_E \leq 3$ and the exact rational certificates bound the optimal two-query

TABLE III. Balanced four-bit tasks grouped by (D, Q_E). Orbits are taken up to coordinate permutations, input complements, and answer exchange; the columns sum to 58 orbits and 6435 partitions. The last column names the source of the lower bound on Q_E: Proposition 4 excludes one query, and the rational dual certificates of Eq. (16) exclude two.

(D, Q_E)	Orbits	Partitions	Representative	Lower bound
$(1,1)$	1	4	b_1	—
$(2,1)$	1	6	$b_1 \oplus b_2$	—
$(2,2)$	1	48	$\mathrm{Sel}(b_1, b_2, b_3)$	Prop. 4
$(3,2)$	8	248	$\mathrm{Maj}(b_1, b_2, b_3)$	Prop. 4
$(4,2)$	4	89	Par_4	Prop. 4
$(3,3)$	10	1536	$\mathrm{Sel}(b_1, b_2 b_3, b_2 \vee b_4)$	certificate
$(4,3)$	33	4504	$b_1 \oplus b_2 \oplus b_3 b_4$	certificate

worst-case success by values between 0.9001 and 0.9862, hence $Q_E = 3$. The classification of Table III is therefore exact: 4847 balanced four-bit partitions have an exact quantum advantage (6 by a factor of two at $(2,1)$, 89 by a factor of two at $(4,2)$, 248 by a factor of $3/2$, and 4504 by a factor of $4/3$) and 1588 have none. The polynomial method cannot give these lower bounds: the 43 orbits have real degree 3 (2144 partitions) or 4 (3896 partitions), so it yields only $Q_E \geq 2$.

Ref. [15] reported numerical SDP values for all Boolean functions of up to four bits and noted that such values do not by themselves prove exactness; later work on exact two-query algorithms builds on that classification [21]. We checked our orbit representatives against the published NPN (negation–permutation–negation) identifiers of Ref. [15]. After removing dummy variables, its one- and two-query entries agree with all 15 representatives settled by trees, and for the other 43 its optimal two-query success probabilities, 0.900 to 0.986, agree with the optimal values 0.9000–0.9861 of our dual solutions. The certificates turn this numerical agreement into exact lower bounds; the 15 orbits settled by trees give dual value one, as they must.

Lean formalization. The Supplemental Material includes a Lean 4/mathlib formalization of the exact algebraic and finite parts. Three `native_decide` checks reproduce the 35-task and 6435-task censuses of decision trees and restricted trees, including the 395 cases settled by trees; `native_decide` also trusts Lean's compiler, through an extra axiom that `#print axioms` reports. The 43 dual certificates are checked by the separate exact verifier, not in Lean. Density operators, general quantum query algorithms, and the general parity lower bound are not formalized.

The census arranges balanced functions as answer partitions, identifies only partitions related by symmetries of the oracle, and attaches a short exact certificate to every class. Its proportions describe small cases only; they are not a criterion for asymptotic speedup.

V. DECOHERED ACCESS: BERNSTEIN–VAZIRANI WITH AN INACCESSIBLE WITNESS

For $a, x \in \mathbb{F}_2^n$, let $f_a(x) = a \cdot x \pmod 2$ and $O_a |x, y\rangle = |x, y \oplus a \cdot x\rangle$. With ordinary phase kickback the states for different a are orthogonal, so one query identifies a [31]. We now let the oracle also leave an inaccessible record of the address, without changing any value it returns on basis states.

Choose $S \subseteq \{1, \ldots, n\}$ with $|S| = k$, let x_S be the corresponding substring of x, and let $\bar{S}$ be the complement of S. Each call acts as

$$W_a^S |x, y, e\rangle = |x, y \oplus a \cdot x, e \oplus x_S\rangle . \tag{18}$$

on a fresh witness register in $|0^k\rangle$, which is then discarded. The oracle the algorithm sees is therefore the channel

$$\Phi_a^S(\rho) = \mathrm{Tr}_E\left[W_a^S(\rho \otimes |0^k\rangle\langle 0^k|)(W_a^S)^\dagger\right] = O_a\, \Delta_S(\rho)\, O_a^\dagger, \tag{19}$$

where $\Delta_S(\rho) = \sum_s \Pi_s \rho \Pi_s$ dephases the address register in the coordinates S (Π_s projects onto $x_S = s$); copying x_S into a discarded register is exactly this dephasing, and O_a commutes with every Π_s. Basis queries still return $a \cdot x$ exactly, but the witness destroys coherence between addresses with different x_S, as in which-way experiments [22] and in the coherence analysis of the Bernstein–Vazirani algorithm [23].

Surviving partition. Define

$$\Pi_S = \{C_u : u \in \mathbb{F}_2^{n-k}\}, \qquad C_u = \{a : a_{\bar{S}} = u\}. \tag{20}$$

For the classical comparison, let $\mathcal{O}^{\mathrm{val}} = \{O_a^{\mathrm{val}} : a \in \mathbb{F}_2^n\}$ be the value oracle obtained by preparing and reading out Φ_a^S in the computational basis: a query chooses x and returns $a \cdot x$. For $k < n$ and worst-case error $\epsilon < 1/2$,

$$Q_\epsilon^\Phi(\Pi_S) = 1, \qquad R_\epsilon(\Pi_S; \mathcal{O}^{\mathrm{val}}) = n - k. \tag{21}$$

For the quantum part, the usual Bernstein–Vazirani input and phase kickback, followed by Eq. (19), leave on

the n-qubit address register (after factoring out the answer qubit, which stays in $|-\rangle$) the state

$$\rho_a^S = 2^{-n} \sum_{x,x' :\, x_S = x'_S} (-1)^{a\cdot(x\oplus x')} \, |x\rangle\langle x'| \,. \tag{22}$$

After the final Hadamard gates, the orthogonality of characters gives, with the coordinates reordered,

$$\begin{aligned} H^{\otimes n} \rho_a^S H^{\otimes n} &= |a_{\bar S}\rangle\langle a_{\bar S}| \otimes \frac{I_S}{2^k}, \\ \Pr(z|a) &= 2^{-k}\, \mathbf{1}[z_{\bar S} = a_{\bar S}]. \end{aligned} \tag{23}$$

So one query resolves Π_S exactly; zero queries cannot, since the output would not depend on the cell. Classically, the $n-k$ basis queries outside S suffice. For the classical lower bound, grant even the promise $a_S = 0$ and let $a_{\bar S}$ be uniformly distributed. Fewer than $n-k$ linear equations leave at least two equally likely candidates, so a decoder that sees the whole transcript succeeds with probability at most $1/2$. Averaging over the protocol's random choices does not change this, so some input has success at most $1/2$.

Identifying a. The same channel also changes the cost of the original task, identifying a.

Proposition 6 (Witness cost of identification)**.** *For $0 \le k \le n$, with S known to the algorithm, the exact quantum cost of identifying a through $\{\Phi_a^S\}$ is $Q_E^{\Phi}(\Pi_{\text{single}}) = \min\{k+1, n\}$; classically, $R_\epsilon(\Pi_{\text{single}}; \mathcal{O}^{\text{val}}) = n$ for $\epsilon < 1/2$.*

Proof. Upper bound: one Bernstein–Vazirani query with $x_S = 0$ returns $a_{\bar S}$, and k basis queries $x = e_i$, $i \in S$, return a_S; alternatively, n basis queries return a. At $k = 0$ this is the ordinary one-query algorithm. Lower bound for $k \ge 1$: record each dephasing in Eq. (19) in an inaccessible environment register. For a record $\vec s = (s_1, \dots, s_T)$, $s_t \in \mathbb{F}_2^k$, let $|\phi_a(\vec s)\rangle$ be the conditional unnormalized output, so that $\rho_a = \sum_{\vec s} |\phi_a(\vec s)\rangle\langle\phi_a(\vec s)|$. Because the records are inaccessible, exact identification requires $\langle\phi_a(\vec s)|\phi_{a'}(\vec s')\rangle = 0$ for all $a \ne a'$ and all records $\vec s, \vec s'$. In the answer qubit's X basis, the tth query multiplies the component at address (s_t, z) either by 1 or by $(-1)^{a\cdot(s_t,z)}$. Hence $\phi_a(\vec s)$ is a linear combination of characters $(-1)^{a\cdot f}$ whose frequencies are $f = 0$ or $f = \sum_{t\in F}(s_t, z_t)$ for nonempty subsets F of the queries. (i) If $s_1, \dots, s_T$ do not span $\mathbb{F}_2^k$, choose $h \ne 0$ orthogonal to all s_t. Every such character is then invariant under $a \mapsto a + (h, 0)$, so $\phi_{a+(h,0)}(\vec s) = \phi_a(\vec s)$, and orthogonality forces $\phi_a(\vec s) = 0$. Hence $T \ge k$, and only records whose s_t span $\mathbb{F}_2^k$ contribute. (ii) Let $T = k < n$. In every contributing record the s_t are linearly independent, so every frequency with $F \ne \emptyset$ has a nonzero S part, and for $u \ne 0$ the sum $\sum_a (-1)^{u\cdot a_{\bar S}} \phi_a(\vec s)$ vanishes. Since the $\phi_a(\vec s)$ are pairwise orthogonal, the squared norm of this sum is $\sum_a \|\phi_a(\vec s)\|^2$, so $\phi_a(\vec s) = 0$ for every contributing record, contradicting normalization. Hence $T \ge k+1$ for $k < n$. The classical bound is the singleton case of the argument above. □

At $k = 0$ nothing is recorded, and Eq. (23) is the ordinary Bernstein–Vazirani output, with one quantum query against n classical ones. Each recorded address bit adds exactly one query until the cost reaches the classical value n; from $k = n-1$ on, the quantum advantage for identification is gone, while the classical cost stays n. If the witness were available to the algorithm, the fixed coupling $|x, e\rangle \mapsto |x, e \oplus x_S\rangle$ could be undone without another oracle call, restoring the ordinary output. What the algorithm accesses here is a channel, not the unitary W_a^S: as with in-place versus standard oracles [4] and faulty oracles [7], whether the oracle leaves an inaccessible record is part of the problem.

VI. DISCUSSION

All results of this paper follow one strategy, which can be stated in four steps; the physics enters in the middle two.

- *Answer partition* Π_g [Eq. (1)]: Put two hidden instances in the same cell whenever the task has the same correct answer for both. The cells C_j are sets of classical instances, not subspaces. The algorithm must learn the cell, and only the cell: in Deutsch's problem the four possible functions fall into two cells, constant and balanced, and the algorithm never has to find out which function within a cell it faces.

- *Oracle access* (Secs. III and V): Fix what one oracle call does. The partition fixes the correct answers to the question; the access model fixes how that answer can be obtained by quantum state discrimination.

- *Probe and one quantum oracle call* (Sec. II): Choose an instance-independent probe state, for instance a uniform superposition obtained by "fanning out" a single basis state with Hadamard gates. One oracle call turns the probe into the output state $|\Psi_a\rangle$ of the hidden instance a.

- *Orthogonality across cells* (Proposition 1): Ask whether some probe makes $\langle\Psi_a|\Psi_{a'}\rangle = 0$ whenever $g(a) \ne g(a')$ [Eq. (5)]. A negative answer needs no search over the infinitely many probes: a single matrix, a dual certificate (Proposition 2), proves that none works. Only orthogonal states can be told apart with certainty, so exactness requires orthogonality, but only between cells. Within a cell, states may overlap or even coincide: the measurement reads the cell label, not the instance, as a post office sorts letters by postal code without reading the street.

Two equally weighted branches of a quantum state cancel, that is, interfere destructively, only if one carries an extra factor -1 relative to the other. In the examples

of the Introduction, the standard flip on a qubit and the transposition on a qutrit both have the eigenvalue -1 and separate constant from balanced functions in one call; the qutrit cyclic shift, whose eigenvalues are the cube roots of unity, cannot (Theorem 1 and Corollary 1). With more than two branches, other phases can cancel, for instance $1+\omega+\omega^2=0$ for $\omega=e^{2\pi i/3}$; this is the interference used by the quantum Fourier transform and Shor's algorithm. Deutsch's problem, however, compares only two address branches, and therefore needs -1. With more calls, the same orthogonality is required of the final states, with instance-independent operations between the calls (Sec. IV); with bounded error, it is relaxed to a required success probability (Proposition 5).

Some of these separations need no non-Clifford operations. The standard Deutsch and Bernstein–Vazirani protocols, including their XOR oracles, use only Clifford operations, and so does the witness coupling of Eq. (18) before the witness is discarded. Given a full classical description of the circuit, they can therefore be simulated efficiently [32, 33]. In the query model, however, such a description would reveal the hidden oracle data, which is more than black-box access provides. For these examples, nonstabilizer resources ("magic" [34]) are not needed for a query advantage. General point oracles with a binary-encoded address need multiply controlled gates, which are not Clifford, so this remark is specific to Deutsch and Bernstein–Vazirani.

The results vary these steps one at a time. Corollary 1 and Proposition 6 change the access (the second step) while holding the task and its classical point values fixed, and the exact cost changes; Proposition 5 shows what survives with bounded error. Theorem 1 and Proposition 4 give the rule for every response unitary. The sign oracle of Proposition 3 is a deliberately weaker interface, whose point values are invisible to basis queries. The census keeps the standard oracle fixed and varies the balanced task (the first step), and the cost changes again; for $N \le 4$ every class is certified exactly. Shor's and Grover's algorithms [35, 36] fit the same picture: each exploits structure of the oracle family that matches a specific coherent access.

Partitions are therefore needed to state a task but do not decide its complexity, and neither do classical point values. A quantum advantage occurs when interference matched to the oracle's structure separates the answer cells in fewer calls than any classical protocol needs with the same access. Superlinear exact separations for total Boolean functions are known [37], but an asymptotic speedup requires a structured family of task–oracle pairs; percentages from a finite census cannot replace that analysis. Within the present framework, the parity question for the qutrit shift (Sec. III) is a natural test of whether a response-spectrum effect persists for all N.

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Data and code availability.—See Supplemental Material at [URL will be inserted by publisher] for the Python and Lean 4 sources, the proof report, the 43 rational dual certificates with an exact verifier, the census tables and witness files, and the cross-validation against the published NPN data of Ref. [15].

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